

# Perfect Absorption with Independent Electric and Magnetic Lattice Resonances in Metallo-Dielectric Arrays

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Lattice resonances are collective modes supported by periodic arrays of nanostructures. They originate from the coherent interaction between the localized modes of the individual constituents of the array, which, for systems made of metallic nanostructures, usually correspond to the electric dipole plasmon. Unfortunately, fundamental symmetry reasons preclude a two-dimensional (2D) arrangement of electric dipoles from absorbing more than half the incident power, thus imposing a strong limitation on the performance of conventional lattice resonances. This work introduces an innovative solution to overcome this constraint, which is based on using an array made of a unit cell containing one metallic and one dielectric nanostructure. Using a rigorous coupled dipole model, it is shown that this system can support two independent lattice resonances associated, respectively, with the electric and magnetic dipole modes of the nanostructures. By adjusting the geometrical characteristics of the array, these two lattice resonances can be meticulously aligned in the spectral domain, leading to the total absorption of the incident power. The results of this work provide clear, yet general, guidelines for the rational design of arrays sustaining lattice resonances capable of producing perfect absorption, thus leveraging the potential of these modes for applications requiring an efficient absorption of light.

electromagnetic modes known as lattice resonances.<sup>[5–7]</sup> Due to their collective nature, lattice resonances are characterized by very strong and spectrally narrow optical responses.<sup>[8–10]</sup> In addition, their resonant wavelengths are commensurate with the array period and, therefore, can be easily tuned by changing the array geometry.<sup>[8–10]</sup> Remarkably, periodic arrays conserve most of the extraordinary abilities of metallic nanostructures to enhance light-matter interactions at the nanoscale,<sup>[6,11,12]</sup> while highly restraining the unwanted nonradiative losses associated with their metallic nature.<sup>[13–15]</sup> As a consequence, arrays sustaining lattice resonances are being utilized to implement biosensors,<sup>[16–18]</sup> light-emitting devices,<sup>[19–21]</sup> optical filters,<sup>[22]</sup> and chiro-optical elements,<sup>[23–26]</sup> among other applications.

A large fraction of the research on lattice resonances has focused on the use of metallic nanostructures much smaller than the wavelength of the light exciting them, for which the interaction between the electric dipoles of the nanostructures is the dominant contribution.<sup>[27–29]</sup>

## 1. Introduction

A promising strategy to attain a narrow optical response using metallic nanostructures consists in arranging them in two-dimensional (2D) periodic arrays.<sup>[1–4]</sup> When the periodicity of the system is larger than the wavelength of the localized plasmons supported by the individual constituents, the coherent multiple scattering between them leads to the excitation of collective

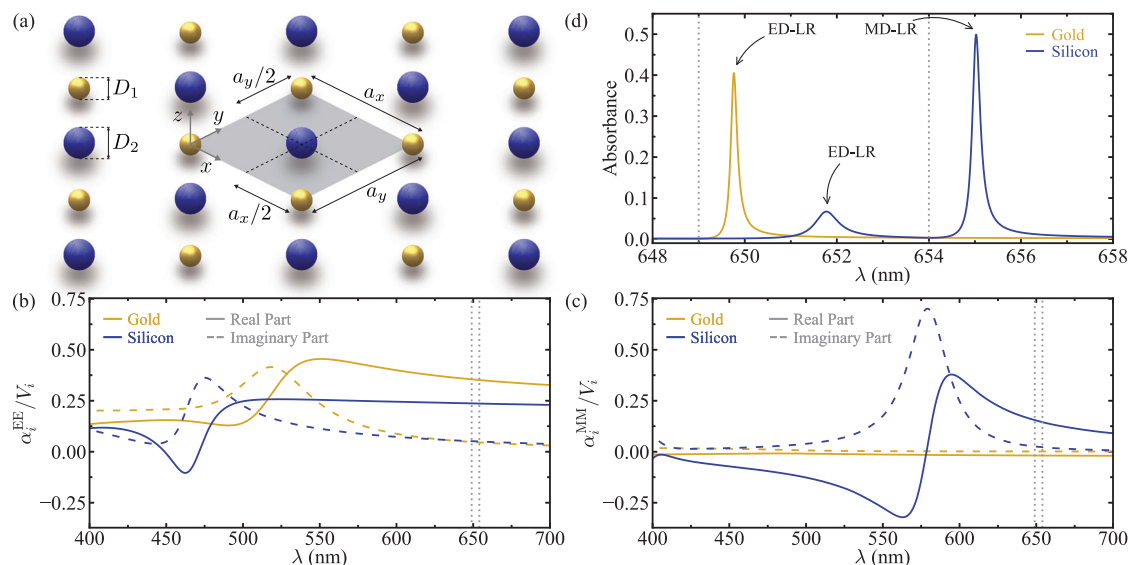
Nevertheless, the interaction between magnetic dipoles, electric and magnetic dipoles, or higher-order multipoles can also give rise to lattice resonances. These couplings become relevant, or even dominant, for arrays built from nanostructures with large sizes, sophisticated morphologies, and/or made of dielectric materials.<sup>[10,30–38]</sup> Importantly, 2D arrays supporting lattice resonances – and metasurfaces in general – in which only electric or magnetic dipoles are involved, are subject to a fundamental limit that prevents them from absorbing more than half of the incident power.<sup>[39]</sup> This restriction hinders the use of lattice resonances for applications requiring an efficient absorption of light, such as light-to-heat conversion,<sup>[40,41]</sup> photodetection,<sup>[42]</sup> and photocatalysis,<sup>[43]</sup> to name a few. One solution to overcome this limit is to place the arrays in the proximity of a perfectly reflective surface, in a configuration known as a “Salisbury screen”,<sup>[39,44,45]</sup> for which perfect absorption can be achieved if the separation between mirror and array is adequately chosen.<sup>[46]</sup> However, this configuration completely blocks the transmission of light at all wavelengths, which can be detrimental for applications in which perfect absorption is required only for a certain wavelength range while perfect transmission is desired for the rest.

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**Figure 1.** a) Schematics of a bipartite array made of gold and silicon nanospheres of diameters  $D_1$  and  $D_2$ . The unit cell (shaded area) is a rectangle with sides of length  $a_x$  and  $a_y$ , with the gold and silicon nanoparticles located, respectively, at the origin ( $x_1 = 0$ ,  $y_1 = 0$ ) and at the center ( $x_2 = a_x/2$ ,  $y_2 = a_y/2$ ) of it. The array is placed in the  $xy$ -plane, surrounded by vacuum, and excited by a normally incident plane wave with the electric and magnetic fields polarized along the  $x$ - and  $y$ -axis, respectively. b) Real (solid lines) and imaginary (dashed lines) parts of the electric polarizabilities of gold (yellow curves) and silicon (blue curves) nanoparticles with diameters  $D_1 = 98$  nm and  $D_2 = 140$  nm. In both cases, we normalize the polarizability to the volume of the nanoparticle  $V_i$ . c) Same as (b) but for the magnetic polarizabilities. d) Absorbance spectra for monopartite rectangular arrays with periods  $a_x = 654$  nm and  $a_y = 649$  nm made of either gold nanoparticles of diameter  $D_1 = 98$  nm (yellow curve) or silicon nanoparticles of diameter  $D_2 = 140$  nm (blue curve). The vertical gray dotted lines in panels (b,c,d) mark the position of the first Rayleigh anomalies of a rectangular array with periods  $a_x = 654$  nm and  $a_y = 649$  nm.

One possible approach to obtaining perfect absorption in a purely 2D system without using a back-reflector consists in making two different resonances overlap spectrally. These resonances must correspond to multipoles of different parity, in what is regarded as a generalized Kerker effect.<sup>[39,47–50]</sup> Notably, these two resonances do not need to be both collective lattice resonances. In fact, most of the previous works on perfect absorption rely on the overlap of a collective lattice resonance of the array with a localized mode supported by its individual constituents.<sup>[51–54]</sup> This procedure, however, has the limitation that the wavelength at which the overlap occurs cannot be easily modified. The reason is that a change in the spectral position of the localized mode can only be accomplished by modifying the polarizability of the nanostructures, which, in turn, modifies the properties of the lattice resonance. Furthermore, this approach usually results in low quality factors due to the broad profile of the localized mode of the individual constituents.

Here, we report a novel approach to attain perfect absorption in planar systems based on the spectral overlap of two different collective lattice resonances, which circumvents the deficiencies of traditional strategies. Our approach relies on the use of a bipartite rectangular array built from the repetition of a unit cell composed of one metallic and one dielectric nanostructure. Arrays with multiple particles per unit cell have been the subject of extensive research<sup>[5,33,55–62]</sup> because, due to their more involved optical response, they provide remarkable flexibility for selectively adjusting the optical properties and spectral position of their lattice resonances.<sup>[5,63]</sup> The particular configuration considered in this work ensures the emergence of two lattice resonances, one

with magnetic and the other with electric dipolar character, whose properties are tuned independently by adjusting the periodicity of the array and the size of the nanostructures. Furthermore, we show that an appropriate choice of these parameters not only leads to perfect absorption but also to higher quality factors than those achieved in previous works using a single lattice resonance. We base our study on the use of a rigorous coupled dipole model (CDM), whose semi-analytic nature allows us to dissect the response of the system and show that the absorption is evenly distributed among the nanostructures and the electric and magnetic dipoles induced in them. This work provides a general framework to design and implement complex 2D arrays capable of sustaining lattice resonances with perfect absorption.

## 2. Results and Discussion

The perfect absorber proposed in this work is schematically depicted in **Figure 1a**. We consider a bipartite periodic rectangular array with periods  $a_x$  and  $a_y$  along the  $x$ - and  $y$ -axis. The array is built from the repetition of a unit cell containing a nanosphere of gold, with diameter  $D_1 = 98$  nm, located in the origin ( $x_1 = 0$ ,  $y_1 = 0$ ), and a nanosphere of silicon, with diameter  $D_2 = 140$  nm, placed at the center of the unit cell ( $x_2 = a_x/2$ ,  $y_2 = a_y/2$ ). This particular arrangement allows for a simple realization of the perfect absorber since, as we discuss below, provides the adequate interactions between the particles in the unit cell. We use gold and silicon nanoparticles because the metallic nature of the former results in a strong electric dipole resonance in the visible part of the spectrum, whereas the high-refractive-index

dielectric nature of the latter produces both electric and magnetic dipole resonances in this same spectral region.<sup>[64,65]</sup> This is demonstrated in Figure 1b,c, where we plot the electric  $\alpha_i^{\text{EE}}$  and magnetic  $\alpha_i^{\text{MM}}$  polarizabilities of the two nanoparticles, normalized to their volumes  $V_i = \pi D_i^3/6$ . These polarizabilities are computed from the dipolar Mie scattering coefficients,<sup>[66]</sup> using the dielectric functions tabulated in Refs. [67] and [68]. As anticipated, the gold nanoparticle only shows an electric dipole resonance around 520 nm, which corresponds to its localized plasmon mode, while the silicon nanoparticle displays both electric and magnetic dipole resonances around 475 and 580 nm, respectively, which are associated with its lowest-order Mie modes.

We assume that the array is located in the  $xy$ -plane, surrounded by vacuum, and excited at normal incidence by an electromagnetic plane wave with the electric and magnetic components polarized along the  $x$ - and  $y$ -axis, respectively. We calculate the optical response of the array using the well-established coupled dipole model (CDM).<sup>[5,26,28,59,69–71]</sup> This approach, which is explained in detail in the Supporting Information, produces accurate results when the period of the array and the wavelength of the electromagnetic field are significantly larger than the size of the nanoparticle. Furthermore, in contrast to fully numerical approaches, the CDM allows us to get the necessary insight to design an array capable of achieving perfect absorption. Within this theoretical framework, the nanoparticles are modeled as point dipoles with both electric and magnetic components, and the collective response of the array is the result of the mutual and self-consistent interaction between the dipoles.

As stated in the introduction, our goal is to achieve perfect absorption by spectrally overlapping the lowest-order electric and magnetic dipole lattice resonances sustained by the bipartite array, which we denote as ED-LR and MD-LR in what follows. This requires a careful choice of the value of  $a_x$  and  $a_y$ . In particular, to obtain strong collective lattice resonances, it is necessary that the first Rayleigh anomalies, which for a rectangular array under normal incidence excitation are located at  $\lambda = a_x$  and  $\lambda = a_y$ , appear at wavelengths larger than those of the electric and magnetic resonances of the individual nanoparticles.<sup>[7,72]</sup> Here, we chose  $a_x = 654$  nm and  $a_y = 649$  nm. These values fulfill the requirement stated above (see Figure 1b,c) and, as we show later, allow us to obtain perfect absorption.

To demonstrate the excitation of lattice resonances under these conditions, in Figure 1d, we plot the absorbance spectra of the two monopartite rectangular arrays, into which the bipartite array under study can be decomposed, for the values of  $a_x$  and  $a_y$  selected above. The monopartite array of gold nanoparticles (yellow curve) sustains a single lattice resonance centered at  $\lambda \gtrsim a_y$ , while for the array of silicon nanoparticles (blue curve), we observe two distinct resonances located at  $\lambda \gtrsim a_y$  and  $\lambda \gtrsim a_x$ . The semi-analytic character of the CDM allows us to determine the nature of these resonances by considering only the electric or the magnetic response of the nanoparticles. By doing so, we conclude that the resonances that appear at  $\lambda \gtrsim a_y$  correspond to ED-LRs, while the one at  $\lambda \gtrsim a_x$  is a MD-LR. This is consistent with the fact that the position of a lattice resonance is controlled by the period along the direction perpendicular to the field component that excites it.

Following the hybridization approach developed in Reference [5], we can construct the response of the bipartite array from

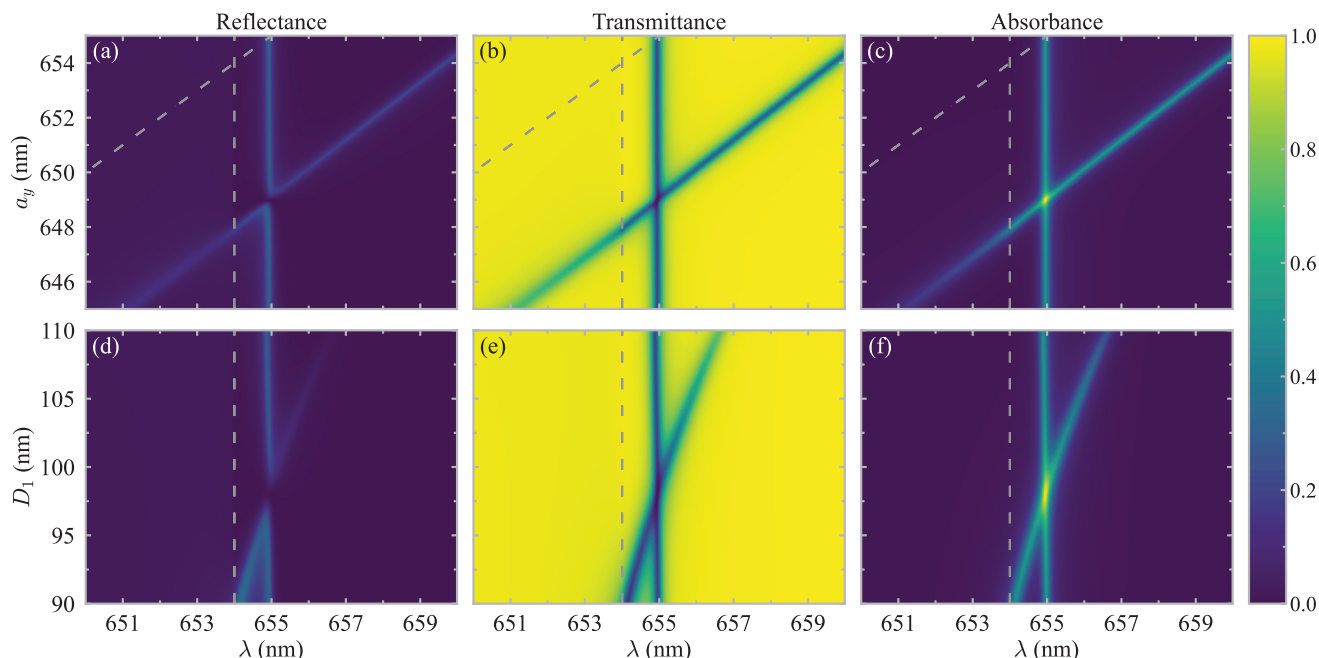
the combination of the responses of the two monopartite arrays analyzed above. Specifically, the optical properties of the lattice resonances of the bipartite array can be determined from the analysis of the following eigenvalue problem (see the Supporting Information for a detailed derivation)

$$\begin{pmatrix} \mathbf{p}_i \\ \mathbf{m}_i \end{pmatrix} = \sum_{j=1}^2 \begin{pmatrix} \alpha_i^{\text{EE}} \mathbb{1}_{3 \times 3} & 0 \\ 0 & \alpha_i^{\text{MM}} \mathbb{1}_{3 \times 3} \end{pmatrix} \begin{pmatrix} \mathcal{G}_{ij}^{\text{EE}} & \mathcal{G}_{ij}^{\text{EM}} \\ \mathcal{G}_{ij}^{\text{ME}} & \mathcal{G}_{ij}^{\text{MM}} \end{pmatrix} \begin{pmatrix} \mathbf{p}_j \\ \mathbf{m}_j \end{pmatrix} \quad (1)$$

Here,  $\mathbf{p}_i$  and  $\mathbf{m}_i$  are the electric and magnetic dipoles induced in the  $i$ -th particle of the unit cell,  $\mathbb{1}_{3 \times 3}$  is the  $3 \times 3$  identity matrix, and  $\mathcal{G}_{ij}^{\zeta}$  represents the different components of the lattice sum. This quantity, which is the Fourier transform of the dipole–dipole interaction tensor, contains all of the information regarding the interaction between the elements of the array. In particular,  $\mathcal{G}_{21}^{\zeta}$  (and equivalently  $\mathcal{G}_{12}^{\zeta}$ ) describes the interaction between the two monopartite arrays.

We aim to independently tune the ED-LR and MD-LR of the bipartite array; that is, to vary the spectral position of the ED-LR without affecting the spectral position of the MD-LR, and vice versa. To that end, we need to place the two particles in the unit cell in a configuration for which the electric-magnetic interaction between them, which is encoded in  $\mathcal{G}_{21}^{\text{EM}}$  and  $\mathcal{G}_{21}^{\text{ME}}$ , vanishes. As shown in Figure S1 (Supporting Information), where we plot the value of the relevant components of  $\mathcal{G}_{21}^{\zeta}$  as a function of the relative position of the nanoparticles in the unit cell, this can be accomplished by placing one particle at the origin of the unit cell and the other one at its center, i.e., following the arrangement described in Figure 1a. Furthermore, for this particular configuration,  $\mathcal{G}_{21}^{\text{EE}}$  is maximized in absolute terms. As a result of these interactions, the bipartite array is expected to support a single ED-LR of subradiant nature, which results from the hybridization of the ED-LRs supported by the gold and silicon monopartite arrays,<sup>[59]</sup> and a single MD-LR, corresponding to the mode supported by the silicon monopartite array. Therefore, in addition to achieving independent ED-LR and MD-LR resonances, the subradiant character of the ED-LR provides an additional contribution to maximize the absorbance. Importantly, under these conditions, and with the polarization of the incident field described above, the electric and magnetic dipoles induced in the nanoparticles are expected to be oriented along the  $x$ - and  $y$ -axis, respectively. Consequently, we anticipate the spectral position of the MD-LR to be governed by  $a_x$ , whereas that of the ED-LR by  $a_y$ .

Figure 2a–c displays the reflectance, transmittance, and absorbance spectra of the bipartite array as a function of  $a_y$ , for  $a_x = 654$  nm,  $D_1 = 98$  nm, and  $D_2 = 140$  nm. As expected by our previous analysis, this array supports only two lattice resonances in the spectral range of interest. The first one, which corresponds to the MD-LR, is centered at  $\lambda \approx 655$  nm, right above the corresponding Rayleigh anomaly indicated with vertical dashed lines. The second lattice resonance is the ED-LR and its spectral position is connected to the Rayleigh anomaly associated with the period  $a_y$ , which is marked by the oblique dashed lines. When the two lattice resonances are spectrally separated, each one produces an absorbance close to the fundamental limit of 0.5, with a quality factor above  $2 \times 10^3$ . However, when  $a_y = 649$  nm, the ED-LR and MD-LR completely overlap and the array displays a near-unity absorbance, with the reflectance and

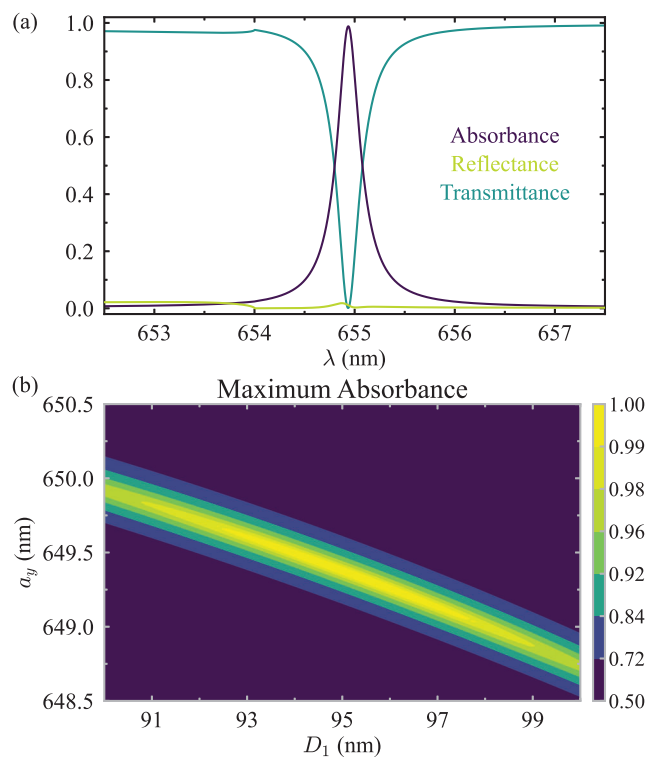


**Figure 2.** Reflectance (a,d), transmittance (b,e), and absorbance (c,f) spectra of the bipartite array. Panels (a–c) are plotted as a function of  $a_y$ , for  $a_x = 654$  nm,  $D_1 = 98$  nm, and  $D_2 = 140$  nm, while panels (d–f) are plotted as a function of  $D_1$ , for  $a_x = 654$  nm,  $a_y = 649$  nm, and  $D_2 = 140$  nm. In all cases, the gray dashed lines indicate the position of the first Rayleigh anomalies of the array.

transmittance dropping practically to zero. These results confirm that, under this configuration, the bipartite array supports two distinct lattice resonances that can be tuned independently from one another by simply changing the period, and, when they spectrally overlap, make the system reach perfect absorption.

An alternative path to tune the spectral position of the lattice resonances consists in modifying the size of the nanoparticles. To illustrate this point, in Figure 2d–f, we plot the reflectance, transmittance, and absorbance spectra of the bipartite array as a function of  $D_1$ , for  $a_x = 654$  nm,  $a_y = 649$  nm, and  $D_2 = 140$  nm. We observe that the effect of changing the size of the gold nanoparticle is analogous to that of the period  $a_y$  analyzed in Figure 2a–c, although to produce the same variation in the spectral position of the ED-LR, it is necessary a larger change in  $D_1$ . In particular, starting from the values for which the overlap occurs ( $a_y = 649$  nm and  $D_1 = 98$  nm), a 1 nm shift in the spectral position of the ED-LR requires a  $\approx 8\%$  change in  $D_1$ , compared to the mere  $\approx 0.2\%$  change required in  $a_y$ . Therefore, despite not being as effective as varying the period, modifying the dimensions of the nanoparticles provides an additional and independent mechanism to tune the spectral position of ED-LR and MD-LR, thus facilitating the achievement of perfect absorption.

To gain a deeper insight into the perfect absorber behavior, we analyze in Figure 3a the optical response of the bipartite array with the optimal geometrical parameters found above. In particular, we plot the absorbance, reflectance, and transmittance for the system with  $a_x = 654$  nm,  $a_y = 649$  nm,  $D_1 = 98$  nm, and  $D_2 = 140$  nm. The absorbance (purple curve) reaches a maximum value of 0.988 at  $\lambda = 654.9$  nm and has a quality factor  $Q = 2.4 \times 10^3$ . At that same wavelength, the values of the reflectance (green curve) and the transmittance (cyan curve) are reduced to 0.01 and 0.001, respectively. Importantly, these results, which are calcu-



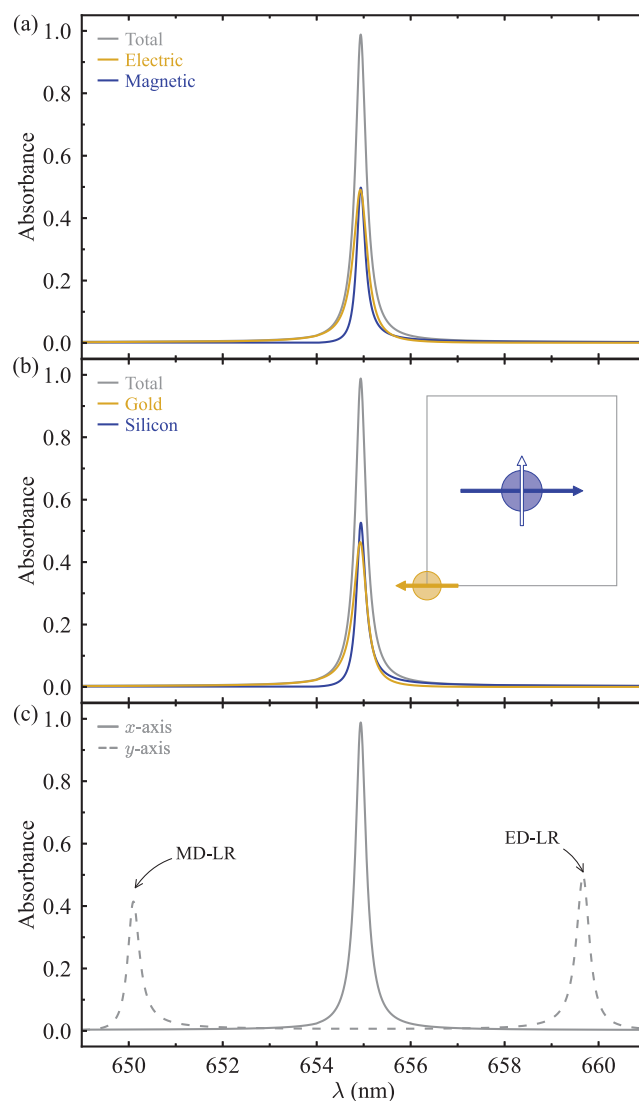
**Figure 3.** a) Absorbance (purple curve), reflectance (green curve), and transmittance (cyan curve) spectra of the bipartite array with  $a_x = 654$  nm,  $a_y = 649$  nm,  $D_1 = 98$  nm, and  $D_2 = 140$  nm. b) Maximum value of absorbance as a function of  $a_y$  and  $D_1$ , assuming  $a_x = 654$  nm and  $D_2 = 140$  nm. Notice that the colorbar follows a logarithmic scale.

lated with the CDM, are in excellent agreement with full solutions of Maxwell's equations obtained using a finite element method (FEM) solver, as demonstrated in Figure S2 (Supporting Information). In addition, in Figure S3 (Supporting Information), we exploit these FEM results to visualize the electric and magnetic field produced by the lattice resonances in the plane of the array.

We can further increase the absorbance of the bipartite array by fine-tuning the geometrical parameters of the system. To illustrate this possibility, in Figure 3b, we plot the maximum value of absorbance as a function of both  $a_y$  and  $D_1$ . We observe that, through an appropriate adjustment of these geometrical parameters, it is possible to reach values of absorbance of up to 0.998. Furthermore, these results show that there is a significant range of periods and diameters that lead to an absorbance above 0.99, suggesting that the choice of parameters is somewhat flexible. In fact, this flexibility can be further increased if we let other parameters of the system, such as the periodicity in the  $x$ -axis  $a_x$  and the size of the silicon nanoparticle  $D_2$ , change as well.

The key element in our approach to attaining perfect absorption is the spectral overlap of the ED-LR and MD-LR supported by the bipartite array. The reasoning is that, under such conditions, each of these modes is able to saturate its fundamental absorption limit,<sup>[39,48]</sup> and provides a contribution of 0.5 to the total absorbance of the array. To verify this behavior, in Figure 4a, we explicitly separate the absorbance (gray curve) of the array analyzed in Figure 3a into the contributions of the electric (yellow curve) and magnetic (blue curve) dipoles. To isolate these contributions, we neglect the magnetic and the electric response of the nanoparticles, respectively, by imposing  $\alpha_i^{MM} = 0$  and  $\alpha_i^{EE} = 0$  in the CDM calculations. As expected, these results confirm that, under perfect absorption conditions, the ED-LR and MD-LR are perfectly tuned to the same wavelength and each provides an almost identical contribution slightly below 0.5 to the overall absorbance of the bipartite array.

It is also insightful to separate the contribution to the absorbance of each of the two nanoparticles in the unit cell. We perform this analysis in Figure 4b, where the yellow and blue curves show, respectively, the absorbance corresponding to the gold and silicon nanoparticles. We observe that, although at first glance both nanoparticles display a similar absorbance, the contribution of the silicon nanoparticle, which reaches a maximum of 0.53, is slightly larger than that of the gold nanoparticle, which remains at 0.46. We attribute this small difference to the fact that, while the gold nanoparticle only sustains the ED-LR, the silicon nanoparticle contributes to both the ED-LR and the MD-LR. To corroborate this hypothesis, we analyze the dipoles induced in the nanoparticles at the wavelength of maximum absorbance. The corresponding results are displayed in the inset of Figure 4b. Specifically, the filled and empty arrows indicate the electric and magnetic dipoles, while the yellow and blue colors correspond to the gold and silicon nanoparticles. In all cases, the length of the arrows is proportional to the imaginary part of the corresponding dipole. As anticipated, the silicon nanoparticle supports both an electric and a magnetic dipole, oriented, respectively, along the  $x$ - and  $y$ -axis. In contrast, the gold nanoparticle only supports an electric dipole that points to the negative  $x$ -axis. These results are consistent with our previous analysis of the different components of the lattice sum, which determine the interaction between the elements of the array. Such analysis



**Figure 4.** a) Absorbance spectra for the bipartite array (gray curve), along with the explicit contributions of electric (yellow curve) and magnetic (blue curve) dipoles. b) Absorbance spectra for the bipartite array (gray curve), along with the contributions of the gold (yellow curve) and the silicon (blue curve) nanoparticles. The inset displays the imaginary part of the electric (filled arrows) and magnetic (empty arrows) dipole moments induced in the gold (yellow) and silicon (blue) nanoparticles, calculated at the wavelength of maximum absorbance. The length of the arrows is proportional to the imaginary part of the corresponding dipole. Notice that the arrow corresponding to the magnetic dipole of the gold nanoparticle is not shown because it is too small to be clearly distinguished. c) Absorbance spectra for the bipartite array when the electric field of the incident plane wave is polarized along the  $x$ - (solid curve) and  $y$ -axis (dashed curve). In all of the cases,  $a_x = 654$  nm,  $a_y = 649$  nm,  $D_1 = 98$  nm, and  $D_2 = 140$  nm.

attributed a subradiant character<sup>[59]</sup> to the ED-LR, arising from the partial cancellation between the electric dipoles of the two nanoparticles, and predicted that the MD-LR was exclusively sustained by the silicon nanoparticles. Therefore, since, as shown in Figure 4a, the MD-LR already contributes with 0.5 to the overall absorbance of the array, we conclude that the electric dipole induced in the silicon nanoparticle is responsible for the small

imbalance in the absorbance of the two nanoparticles in the unit cell.

In the design of the perfect absorber bipartite array, we have assumed from the beginning that the incident electromagnetic plane wave has its electric and magnetic components polarized, respectively, along the  $x$ - and  $y$ -axis. It is then very instructive to examine how the optical response of the system changes when it is excited with the orthogonal polarization. We perform this analysis in Figure 4c for the bipartite array analyzed in Figure 3a. Specifically, the solid and dashed curves show, respectively, the absorbance of the array when the electric field of the incident plane wave is polarized along the  $x$ - and  $y$ -axis. In the former case, the array sustains, as we have shown repeatedly, a single resonance with perfect absorption. However, under the orthogonal excitation, the spectrum displays two different lattice resonances, both of which produce an absorbance close to 0.5. The mode appearing at shorter wavelengths corresponds to the MD-LR, while the one at larger wavelengths is the ED-LR. This behavior is completely expected, as rotating the polarization by 90 degrees is analogous to swapping the periods  $a_x$  and  $a_y$ . Doing so alters the positions of the ED-LR and MD-LR and prevents the system from fulfilling the perfect absorption condition. Therefore, our approach to obtaining perfect absorption in a 2D array is sensitive to the polarization of the incident electromagnetic field. Nevertheless, this sensitivity offers new functionalities, as the system can also work as an almost perfect narrowband linear polarizer, which transmits above 99% of the incoming light for one polarization and only  $\approx 0.1\%$  for the orthogonal one.

### 3. Conclusion

In summary, we have presented a novel strategy to achieve perfect absorption in 2D periodic arrays sustaining collective lattice resonances. Our approach is based on spectrally overlapping the electric (ED-LR) and magnetic (MD-LR) dipole lattice resonances, supported by a thoughtfully designed array of metallic and dielectric nanoparticles. The different symmetry of the radiation patterns of these modes ensures that their interference results in the perfect absorption of all of the incident radiation. Although similar approaches have been explored in the past for the achievement of a generalized Kerker effect,<sup>[47,73–75]</sup> our results are the first to theoretically demonstrate their use to achieve perfect absorption. Furthermore, the use of two lattice resonances provides very high quality factors, which makes our approach stand out above other procedures to achieve perfect absorption.

To illustrate our approach, we have studied a bipartite periodic array built from the repetition of a rectangular unit cell containing one gold and one silicon nanosphere. Using a rigorous coupled dipole model (CDM), we have shown that this array sustains both ED-LR and MD-LR that can be independently tuned by changing the periods and the diameters of the nanoparticles. Among these two tuning mechanisms, we have found that adjusting the period offers a broader range of tunability compared to altering the diameters. This is highly convenient from the point of view of an experimental implementation, as this parameter can be externally controlled if the arrays are embedded in flexible<sup>[76–79]</sup> or other stimuli-responsive<sup>[80]</sup> substrates. Exploiting the insight that the semi-analytic nature of the CDM provides, we have found that, under the conditions of perfect absorption, half

of the incident power is absorbed by the ED-LR, and half by the MD-LR. Moreover, the absorption is evenly distributed between the two nanoparticles in the unit cell, which could be relevant for applications in thermoplasmonics<sup>[41]</sup> and photocatalysis.<sup>[43]</sup> Importantly, we have shown that, given a certain set of parameters for which the ED-LR and MD-LR overlap, the use of an orthogonal polarization causes the detuning of these lattice resonances, thus suppressing the perfect absorption behavior. This sensitivity to the incident polarization can be exploited to obtain a perfect narrowband linear polarizer that perfectly blocks one polarization and completely transmits the orthogonal one. Although for simplicity we have studied an infinite array placed in vacuum, our analysis is directly applicable to any homogeneous dielectric environment by appropriately accounting for the corresponding shift in the spectral position of the lattice resonances. However, a significant difference in the dielectric properties of the media above and below the array can result in the degradation of the collective lattice resonances predicted here.<sup>[81]</sup> Fortunately, there are several strategies already explored that can minimize the impact of this effect. These include, for instance, the use of index-matching fluids and polymers with the appropriate characteristics.<sup>[82]</sup> The results of this work provide the theoretical understanding necessary to achieve perfect absorption using the lattice resonances supported by 2D arrays, and, therefore, are expected to have far-reaching implications from both fundamental and applied perspectives.

### Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

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### Conflict of Interest

The authors declare no conflict of interest.

### Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

### Keywords

Kerker effect, lattice resonances, nanoparticle arrays, non-Bravais lattices, perfect absorption, periodic arrays

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- [1] S. Zou, N. Janel, G. C. Schatz, *J. Chem. Phys.* **2004**, *120*, 10871.
- [2] R. Adato, A. A. Yanik, C.-H. Wu, G. Shvets, H. Altug, *Opt. Express* **2010**, *18*, 4526.
- [3] V. I. Zakomirnyi, I. L. Rasskazov, V. S. Gerasimov, A. E. Ershov, S. P. Polyutov, S. V. Karpov, *Appl. Phys. Lett.* **2017**, *111*, 123107.
- [4] D. Khlopin, F. Laux, W. P. Wardley, J. Martin, G. A. Wurtz, J. Plain, N. Bonod, A. V. Zayats, W. Dickson, D. Gérard, *J. Opt. Soc. Am. B* **2017**, *34*, 691.
- [5] S. Baur, S. Sanders, A. Manjavacas, *ACS Nano* **2018**, *12*, 1618.
- [6] A. Manjavacas, L. Zundel, S. Sanders, *ACS Nano* **2019**, *13*, 10682.
- [7] L. Zundel, J. R. Deop-Ruano, R. Martínez-Herrero, A. Manjavacas, *ACS Omega* **2022**, *7*, 31431.
- [8] W. Wang, M. Ramezani, A. I. Väkeväinen, P. Törmä, J. Gómez Rivas, T. W. Odom, *Mater. Today* **2018**, *21*, 303.
- [9] V. G. Kravets, A. V. Kabashin, W. L. Barnes, A. N. Grigorenko, *Chem. Rev.* **2018**, *118*, 5912.
- [10] A. D. Utyshev, V. I. Zakomirnyi, I. L. Rasskazov, *Rev. Phys.* **2021**, *6*, 100051.
- [11] S. Zou, G. C. Schatz, *Chem. Phys. Lett.* **2005**, *403*, 62.
- [12] A. G. Nikitin, A. V. Kabashin, H. Dallaporta, *Opt. Express* **2012**, *20*, 27941.
- [13] Q. Le-Van, E. Zoethout, E.-J. Geluk, M. Ramezani, M. Berghuis, J. Gómez Rivas, *Adv. Opt. Mater.* **2019**, *7*, 1801451.
- [14] S. Deng, R. Li, J.-E. Park, J. Guan, P. Choo, J. Hu, P. J. M. Smeets, T. W. Odom, *Proc. Natl. Acad. Sci.* **2020**, *117*, 23380.
- [15] M. S. Bin-Alam, O. Reshef, Y. Mamchur, M. Z. Alam, G. Carlow, J. Upham, B. T. Sullivan, J.-M. Ménard, M. J. Huttunen, R. W. Boyd, K. Dolgaleva, *Nat. Commun.* **2021**, *12*, 974.
- [16] R. Adato, A. A. Yanik, J. J. Amsden, D. L. Kaplan, F. G. Omenetto, M. K. Hong, S. Erramilli, H. Altug, *Proc. Natl. Acad. Sci.* **2009**, *106*, 19227.
- [17] A. Danilov, G. Tselikov, F. Wu, V. G. Kravets, I. Ozerov, F. Bedu, A. N. Grigorenko, A. V. Kabashin, *Biosens. Bioelectron.* **2018**, *104*, 102.
- [18] C. Matricardi, C. Hanske, J. L. Garcia-Pomar, J. Langer, A. Mihi, L. M. Liz-Marzán, *ACS Nano* **2018**, *12*, 8531.
- [19] G. Lozano, D. J. Louwers, S. R. K. Rodríguez, S. Murai, O. T. A. Jansen, M. A. Verschuuren, J. Gómez Rivas, *Light Sci. Appl.* **2013**, *2*, e66.
- [20] W. Zhou, M. Dridi, J. Y. Suh, C. H. Kim, D. T. Co, M. R. Wasielewski, G. C. Schatz, T. W. Odom, *Nat. Nanotechnol.* **2013**, *8*, 506.
- [21] R. Guo, M. Nečada, T. K. Hakala, A. I. Väkeväinen, P. Törmä, *Phys. Rev. Lett.* **2019**, *122*, 013901.
- [22] A. Kristensen, J. K. W. Yang, S. I. Bozhevolnyi, S. Link, P. Nordlander, N. J. Halas, N. A. Mortensen, *Nat. Rev. Mater.* **2017**, *2*, 16088.
- [23] E. S. A. Goerlitz, R. Mohammadi, S. Nechayev, K. Volk, M. Rey, P. Banzer, M. Karg, N. Vogel, *Adv. Mater.* **2020**, *32*, 2001330.
- [24] A. Movsesyan, L. V. Besteiro, X.-T. Kong, Z. Wang, A. O. Govorov, *Adv. Opt. Mater.* **2022**, *10*, 2101943.
- [25] M. Manocchio, V. Tasco, F. Todisco, A. Passaseo, M. Cuscuna, I. Tarantini, G. Gigli, M. Esposito, *Adv. Sci.* **2023**, *10*, 2206930.
- [26] L. Cerdán, L. Zundel, A. Manjavacas, *ACS Photonics* **2023**, *10*, 1925.
- [27] F. J. García de Abajo, *Rev. Mod. Phys.* **2007**, *79*, 1267.
- [28] C. Cherqui, M. R. Bourgeois, D. Wang, G. C. Schatz, *Acc. Chem. Res.* **2019**, *52*, 2548.
- [29] L. Zundel, A. Cuartero-González, S. Sanders, A. I. Fernández-Domínguez, A. Manjavacas, *ACS Photonics* **2022**, *9*, 540.
- [30] V. Giannini, G. Vecchi, J. Gómez Rivas, *Phys. Rev. Lett.* **2010**, *105*, 266801.
- [31] V. E. Babicheva, A. B. Evlyukhin, *ACS Photonics* **2018**, *5*, 2022.
- [32] V. E. Babicheva, *MRS Adv.* **2019**, *4*, 713.
- [33] R. Li, M. R. Bourgeois, C. Cherqui, J. Guan, D. Wang, J. Hu, R. D. Schaller, G. C. Schatz, T. W. Odom, *Nano Lett.* **2019**, *19*, 6435.
- [34] G. W. Castellanos, P. Bai, J. Gómez Rivas, *J. Appl. Phys.* **2019**, *125*, 213105.
- [35] S. Murai, G. W. Castellanos, T. V. Raziman, A. G. Curto, J. G. Rivas, *Adv. Opt. Mater.* **2020**, *8*, 1902024.
- [36] X. Zhao, L. Xiong, Z. Zhang, G. Li, *Opt. Express* **2022**, *30*, 34601.
- [37] Y. Li, C. Tan, J. Hu, W. Bai, R. Zhang, Q. Lin, Y. Zhang, L. Wang, *Results Phys.* **2022**, *39*, 105730.
- [38] S. Zhang, Y. Zhang, X. Yin, L. Kong, C. Huang, *J. Appl. Phys.* **2023**, *134*, 053102.
- [39] R. Alaei, M. Albooyeh, C. Rockstuhl, *J. Phys. D* **2017**, *50*, 503002.
- [40] M. R. Bourgeois, A. W. Rossi, S. Khorasani, D. J. Masiello, *ACS Photonics* **2022**, *9*, 3656.
- [41] L. Zundel, K. Malone, L. Cerdán, R. Martínez-Herrero, A. Manjavacas, *ACS Photonics* **2023**, *10*, 274.
- [42] S. V. Zhukovsky, V. E. Babicheva, A. V. Uskov, I. E. Protsenko, A. V. Lavrinenko, *Plasmonics* **2014**, *9*, 283.
- [43] H. Jia, C. C. Tsoi, A. E. Abed, W. Yu, A. Jian, S. Sang, X. Zhang, *Laser Photon. Rev.* **2023**, *17*, 2200700.
- [44] W. W. Salisbury, *Absorbent body for electromagnetic waves*, U.S. Patent 2 599 944 **1952**.
- [45] S. Thongrattanasiri, F. H. L. Koppens, F. J. García de Abajo, *Phys. Rev. Lett.* **2012**, *108*, 047401.
- [46] Y. Bai, H. Zheng, Q. Zhang, Y. Yu, S. Liu, *Opt. Express* **2022**, *30*, 45400.
- [47] V. E. Babicheva, A. B. Evlyukhin, *Laser Photon. Rev.* **2017**, *11*, 1700132.
- [48] W. Liu, Y. S. Kivshar, *Opt. Express* **2018**, *26*, 13085.
- [49] R. Dezert, P. Richetti, A. Baron, *Opt. Express* **2019**, *27*, 26317.
- [50] M. R. Bourgeois, A. W. Rossi, M. Chalifour, C. Cherqui, D. J. Masiello, *J. Phys. C* **2021**, *125*, 18817.
- [51] C.-Y. Yang, J.-H. Yang, Z.-Y. Yang, Z.-X. Zhou, M.-G. Sun, V. E. Babicheva, K.-P. Chen, *ACS Photonics* **2018**, *5*, 2596.
- [52] L. Xiong, H. Ding, Y. Lu, G. Li, *J. Phys. D* **2022**, *55*, 185106.
- [53] Y.-T. Bai, H.-Y. Zheng, Q. Zhang, S.-D. Liu, *Opt. Laser Technol.* **2022**, *156*, 108460.
- [54] Z.-X. Zhou, M.-J. Ye, M.-W. Yu, J.-H. Yang, K.-L. Su, C.-C. Yang, C.-Y. Lin, V. E. Babicheva, I. V. Timofeev, K.-P. Chen, *ACS Nano* **2022**, *16*, 5994.
- [55] P. Lunnemann, A. F. Koenderink, *Phys. Rev. B* **2014**, *90*, 245416.
- [56] A. D. Humphrey, N. Meinzer, T. A. Starkey, W. L. Barnes, *ACS Photonics* **2016**, *3*, 634.
- [57] A. D. Humphrey, W. L. Barnes, *J. Opt.* **2016**, *18*, 035005.
- [58] I. M. Fradkin, S. A. Dyakov, N. A. Gippius, *Phys. Rev. B* **2020**, *102*, 045432.
- [59] A. Cuartero-González, S. Sanders, L. Zundel, A. I. Fernández-Domínguez, A. Manjavacas, *ACS Nano* **2020**, *14*, 11876.
- [60] L. Zundel, A. May, A. Manjavacas, *ACS Photonics* **2021**, *8*, 360.
- [61] A. Warren, M. M. Alkaisi, C. P. Moore, *J. Vac. Sci. Technol.* **2021**, *39*, 063601.
- [62] S. Murai, D. R. Abujetas, L. Liu, G. W. Castellanos, V. Giannini, J. A. Sánchez-Gil, K. Tanaka, J. Gómez Rivas, *Laser Photon. Rev.* **2022**, *16*, 2100661.
- [63] M. Braïk, T. Geronimi-Jourdain, S. Lau-Truong, A. Belkhir, S. Gam-Derouich, A. Chevillat-Biraud, C. Mangeney, N. Félidj, *Nanophotonics* **2023**, *12*, 2179.
- [64] A. García-Etxarri, R. Gómez-Medina, L. S. Froufe-Pérez, C. López, L. Chantada, E. Scheffold, J. Aizpurua, M. Nieto-Vesperinas, J. J. Sáenz, *Opt. Express* **2011**, *19*, 4815.
- [65] A. B. Evlyukhin, S. M. Novikov, U. Zywiets, R. L. Eriksen, C. Reinhardt, S. I. Bozhevolnyi, B. N. Chichkov, *Nano Lett.* **2012**, *12*, 3749.
- [66] F. J. García de Abajo, *Phys. Rev. B* **1999**, *60*, 6086.
- [67] R. L. Olmon, B. Slovick, T. W. Johnson, D. Shelton, S.-H. Oh, G. D. Boreman, M. B. Raschke, *Phys. Rev. B* **2012**, *86*, 235147.
- [68] C. Schinke, P. Christian Peest, J. Schmidt, R. Brendel, K. Bothe, M. R. Vogt, I. Kröger, S. Winter, A. Schirmacher, S. Lim, H. T. Nguyen, D. MacDonald, *AIP Adv.* **2015**, *5*, 067168.
- [69] T. V. Teperik, A. Degiron, *Phys. Rev. B* **2012**, *86*, 245425.
- [70] R. Kolkowski, A. F. Koenderink, *Proc. IEEE* **2020**, *108*, 795.

- [71] L. Zundel, A. Manjavacas, *J. Phys.: Photonics* **2019**, 1, 015004.
- [72] J. R. Deop-Ruano, S. Sanders, A. Alabastri, W. J. M. Kort-Kamp, D. A. R. Dalvit, A. Manjavacas, *Phys. Rev. Applied* **2022**, 18, 044071.
- [73] L. Xiong, X. Luo, H. Ding, Y. Lu, G. Li, *J. Phys. D* **2022**, 55, 395107.
- [74] V. S. Gerasimov, A. E. Ershov, R. G. Bikbaev, I. L. Rasskazov, I. L. Isaev, P. N. Semina, A. S. Kostyukov, V. I. Zakomirnyi, S. P. Polyutov, S. V. Karpov, *Phys. Rev. B* **2021**, 103, 035402.
- [75] L. Xiong, X. Zhao, X. Du, S. Chen, Y. Lu, H. Ding, G. Li, *Nano Res.* **2023**, 16, 3195.
- [76] A. Yang, A. J. Hryn, M. R. Bourgeois, W.-K. Lee, J. Hu, G. C. Schatz, T. W. Odom, *Proc. Natl. Acad. Sci.* **2016**, 113, 14201.
- [77] K.-H. Chang, J.-S. Cheng, T.-W. Lu, P.-T. Lee, *Opt. Express* **2018**, 26, 33215.
- [78] V. Gupta, P. T. Probst, F. R. Goßler, A. M. Steiner, J. Schubert, Y. Brasse, T. A. F. König, A. Fery, *ACS Appl. Mater. Interfaces* **2019**, 11, 28189.
- [79] M. Charconnet, C. Kuttner, J. Plou, J. L. García-Pomar, A. Mihi, L. M. Liz-Marzán, A. Seifert, *Small Methods* **2021**, 5, 2100453.
- [80] O. A. M. Abdelraouf, Z. Wang, H. Liu, Z. Dong, Q. Wang, M. Ye, X. R. Wang, Q. J. Wang, H. Liu, *ACS Nano* **2022**, 16, 13339.
- [81] B. Auguié, X. M. Bendaña, W. L. Barnes, F. J. García de Abajo, *Phys. Rev. B* **2010**, 82, 155447.
- [82] K. Volk, J. P. S. Fitzgerald, M. Karg, *ACS Appl. Mater. Interfaces* **2019**, 11, 16096.